

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: -6 ; $+4$
- Ex 1b: 9 , -15 , 0
- Ex 2a: -3 , $+3$; back where it started
- Ex 2b: -8 (8 units from 0)
- Ex 3a: -25
- Ex 3b: 7 and -7

Easy • Set A (quick drill)

- -5
- 40
- -12
- 8
- -3
- 200
- 8
- -12
- 25
- 0
- -40
- 1

Easy • Set B

- 2
- 23
- -56
- 10
- -8
- 0

Medium • Set C

- 3 steps back; $+3$ and -3
- $+14$ m (14 m above sea level)
- -7
- -12 (12 units from 0)
- 5 and -5
- You owe \$60 (overdrawn)

Challenging • Set D

- No, $-0 = 0$
- -18
- -2 , -1 , 0 , 1 , 2
- Yes: every integer has an opposite (0 is its own)

Extension

- 11 ; -11
- No: if n is negative, $-n$ is positive (e.g. $n = -3$, $-n = 3$)

Exit ticket

- Level 1: 15
- Level 2: -9 ; 9
- Level 3: 31 , because the opposite of 31 is -31

1.02 The number line (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: dots at -5 , -2 , 0 , 3
- Ex 1b: -2 , -1 , 0 , 1
- Ex 2a: dots at -4 , -1 , 2 , 5
- Ex 2b: -5
- Ex 3a: scale in 5s; dots at -30 , -15 , 10
- Ex 3b: 9

Easy • Set A

- -7
- -3
- 0
- 2
- 5
- -1

Easy • Set B

- dots at -3 , -1 , 1 , 5 , 6
- dots at -7 , -4 , 0 , 4

Medium • Set C

- -1 , 0 , 1 , 2 , 3 , 4
- -5 , -4 , -3 , -2 , -1 , 0
- -7 , -6 , -5 , -4
- -6 , -4 , -2 , 0 , 2 , 4 , 6

Challenging • Set D

- scale -40 to 40 in 10s with dots
- -3 , -2 , -1 , 0 , 1 , 2
- 17
- -3

Extension

- 54
- -6

Exit ticket

- Level 1: -2 , -1 , 0
- Level 2: -8
- Level 3: -3

1.03 Comparing and ordering integers

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: False; false
- Ex 1b: -6 , -1 , 0 , 3 , 5
- Ex 2a: $>$
- Ex 2b: 8 , 3 , 0 , -2 , -15
- Ex 3a: -5 , $-(2)$, 1 , $-(-3)$
- Ex 3b: -11 , -10 , -9

Easy • Set A (quick drill)

- F
- F
- T
- T
- T
- F
- $<$
- $<$
- $>$
- $>$
- $>$
- $<$
- $>$
- $>$
- $<$

Easy · Set B

1. >
2. <
3. >
4. >
5. <
6. <

Medium · Set C

1. -8, -2, 0, 4, 7
2. 14, 3, 0, -1, -5, -9, -10
3. -20, -15, -3, 8, 12
4. 6, -1, -2, -7, -11
5. -12, -4, -1, 0, 3
6. 40, 12, -8, -35

Challenging · Set D

1. e.g. -6, -7, -8
2. -8
3. -1000, -100, -10, -1, 0
4. Yes, for negatives: -20 is smaller than -2

Extension

1. -4, -(-2), 3, -(-4), -(-9)
2. e.g. 0 and 2 (any integer from -3 to 2)

Exit ticket

1. Level 1: False
2. Level 2: 2, 0, -1, -5
3. Level 3: -20, -12, 0, 5, -(-8)

1.04 Adding integers (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: -1
2. Ex 1b: -10; 4
3. Ex 2a: -11
4. Ex 2b: 1
5. Ex 3a: 5°C
6. Ex 3b: -13

Easy · Set A (quick drill)

1. 3
2. 4
3. -7
4. -3
5. -10
6. 3
7. 3
8. -13
9. -4
10. 6
11. -7
12. -5
13. -9
14. 7
15. -16
16. -7

Easy · Set B

1. -5
2. 0
3. -15
4. 14
5. -5
6. -14

Medium · Set C

1. -11
2. 1
3. -8
4. 4
5. -11
6. -10

Challenging · Set D

1. 5°C
2. level 5
3. -\$15
4. 6

Extension

1. -12
2. 7

Exit ticket

1. Level 1: 3
2. Level 2: -7
3. Level 3: 1°C

1.05 Subtracting integers (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: -4; -12
2. Ex 1b: 7; 4
3. Ex 2a: -12
4. Ex 2b: 6
5. Ex 3a: 15°C
6. Ex 3b: -8

Easy · Set A (quick drill)

1. -4
2. -12
3. -7
4. -7
5. -5
6. -20
7. 13
8. 3
9. -11
10. -10
11. 11
12. -7
13. -7
14. 20
15. -18
16. -15

Easy · Set B

1. 7
2. 4
3. 0
4. 10
5. -9
6. 7

Medium · Set C

1. -12
2. -7
3. 6
4. -11
5. -5
6. 0

Challenging · Set D

1. 16°C
2. 25 m
3. -75 m (75 m deep)
4. -13

Extension

1. -8
2. -10

Exit ticket

1. Level 1: -9
2. Level 2: 13
3. Level 3: 19°C

1.06 Multiplying integers (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: -18 ; -10
2. Ex 1b: 48; -24
3. Ex 2a: -42
4. Ex 2b: 40
5. Ex 3a: -8
6. Ex 3b: Positive: 100 negatives make 50 pairs

Easy · Set A (quick drill)

1. -18
2. -10
3. -44
4. -63
5. -24
6. -100
7. 15
8. -14
9. -24
10. 18
11. -30
12. -28
13. 100
14. -24
15. -32
16. 36

Easy · Set B

1. 48
2. 25
3. 21
4. 24
5. 20
6. 36

Medium · Set C

1. -24
2. -42
3. 40
4. 9
5. 100
6. -8

Challenging · Set D

1. -28 m (28 m deep)
2. -7
3. Negative: 2 pairs make positive, 1 negative left
4. -18°C

Extension

1. -32
2. 8 and -3

Exit ticket

1. Level 1: -42
2. Level 2: 72
3. Level 3: -60

1.07 Dividing integers (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 3; -3
2. Ex 1b: 2
3. Ex 2a: -3
4. Ex 2b: 8
5. Ex 3a: $-\$12$
6. Ex 3b: 54

Easy · Set A (quick drill)

1. 3
2. -3
3. 1
4. -4
5. -2
6. -100
7. -4
8. 4
9. -9
10. -5
11. 6
12. -8
13. -5
14. 3
15. -7
16. 9

Easy · Set B

1. -5
2. 7
3. -9
4. 4
5. 0
6. -9

Medium · Set C

1. 2
2. -3
3. 8
4. -4
5. 2
6. 8

Challenging · Set D

1. $-\$21$
2. -3°C each hour
3. -35
4. -8

Extension

1. -60
2. -5

Exit ticket

1. Level 1: -4
2. Level 2: 7
3. Level 3: -4

1.08 Order of operations with integers

(booklet p. 31)

WE DO examples (teacher)

1. Ex 1a: -12
2. Ex 1b: -4
3. Ex 2a: -18
4. Ex 2b: 7
5. Ex 3a: $1 \div (-3) = -1/3$
6. Ex 3b: 0

Easy · Set A (quick drill)

1. -14
2. -13
3. 0
4. 13
5. 18
6. -3
7. 5
8. 17
9. -8
10. 2
11. -7
12. -4
13. -14
14. 7
15. -14
16. 3

Easy · Set B

1. -18
2. -19
3. -15
4. -2
5. 15
6. 2

Medium · Set C

1. 7
2. 0
3. 12
4. 2
5. -1
6. -2

Challenging · Set D

1. -4
2. $-1/3$
3. 4
4. 4

Extension

1. $(8 - 12) \div (-4 + 2) = 2$
2. e.g. $-3 \times 4 - 5 = -17$

Exit ticket

1. Level 1: 1
2. Level 2: 15
3. Level 3: 12

1.09 Integers on a calculator (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: -9 ; -6
2. Ex 1b: 5
3. Ex 2a: -12
4. Ex 2b: 87
5. Ex 3a: -52
6. Ex 3b: 4

Easy · Set A (quick drill)

1. -9
2. -6
3. -83
4. -48
5. 12
6. -75
7. -69
8. -140
9. -900
10. 36
11. -233
12. 399
13. -32
14. 925

Easy · Set B

1. -12
2. -52
3. 87
4. -4
5. -7
6. 169

Medium · Set C

1. 5
2. 9
3. 36
4. 35
5. -3
6. 36

Challenging · Set D

1. ≈ -1000 ; -1029
2. ≈ -20 ; -20.5
3. The calculator squares 6 first, then makes it negative.
Enter $(-6)^2$
4. It should be 8: $\times$ is done before $-$

Extension

1. 5
2. -5

Exit ticket

1. Level 1: -45
2. Level 2: 5
3. Level 3: 5

1.10 Integer problems (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: 16°C
2. Ex 1b: $-\$45$
3. Ex 2a: level -3
4. Ex 2b: $\$85$
5. Ex 3a: $\$820$
6. Ex 3b: -12 m

Easy · Set A

1. 17°C
2. -4°C
3. -3°C
4. -2°C

Easy · Set B

1. $12\text{ m deep } (-11\text{ m})$
2. 5 floors
3. 550 m
4. 155 m

Medium · Set C

1. $-\$25$
2. $\$45$
3. $-\$60$
4. $\$150$

Challenging · Set D

1. $26 \times 80 - 6 \times 210 = \820
2. $35 - 27 = 8$

Extension

1. 70 minutes
2. e.g. It was 4°C and fell 12°C

Exit ticket

1. Level 1: 4°C
2. Level 2: 18 m
3. Level 3: -7